Yue-Hong Chen

Ph.D. Candidate in Astronomy

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YuehongChen
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Research Interests

My research bridges solar and stellar physics. Starting from observational studies of solar eruptive activity, I have extended my work to high-resolution magnetohydrodynamic (MHD) simulations of Sun-like stars, covering the full chain from stellar coronae and winds to coronal mass ejections and the associated energetic particle events.

Education

Sep 2023 – Present **Ph.D. in Astronomy**, *Nanjing University*, Nanjing, China.
Advisor: Prof. Xin Cheng.
Research focus: high-resolution MHD modelling of stellar coronae, winds, and CMEs in Sun-like stars.

Sep 2025 – Present **Visiting Ph.D. Researcher**, *Leibniz Institute for Astrophysics Potsdam (AIP)*, Potsdam, Germany.
Host: Dr. Julián D. Alvarado-Gómez.
Research focus: stellar wind and CME modelling of Sun-like stars at varying rotation rates.

Sep 2021 – Jun 2023 **M.Sc. in Astronomy**, *Nanjing University*, Nanjing, China.
Advisor: Assoc. Prof. Yu Dai.
Research focus: observational study of solar flares and filament eruptions.

Additional Research Visits

Sep – Nov 2024 **Visiting Researcher**, *Leibniz Institute for Astrophysics Potsdam (AIP)*, Potsdam, Germany.

Aug 2023 **Joint-Training Visiting Student**, *Yunnan University*, Kunming, China.

Publications

Refereed Journal Articles

2025 **Chen, Y.-H.**, Alvarado-Gómez, J. D., Cheng, X., Dai, Y., Shi, T., Poppenhäger, K., Xing, C., Inoue, S., Warnecke, J., Korpi-Lagg, M. J., & Ding, M. *High-Resolution Modeling of Coronae and Winds in Solar-Type Stars with Varying Rotation Rates. I. X-Ray Coronae*. *ApJ*, 995, 83. doi:10.3847/1538-4357/ae1697

2023 **Chen, Y.**, Cheng, X., Chen, J., Dai, Y., & Ding, M. *Observations of a Failed Solar Filament Eruption Involving External Reconnection*. *ApJ*, 959, 67. doi:10.3847/1538-4357/ad09d8

2023 **Chen, Y.**, Dai, Y., & Ding, M. *An Atypical Plateau-Like Extreme-Ultraviolet Late-Phase Solar Flare Driven by the Nonradial Eruption of a Magnetic Flux Rope*. *A&A*, 675, A147. doi:10.1051/0004-6361/202345914

Submitted / Under Review

Chen, Y.-H., Alvarado-Gómez, J. D., et al., *High-Resolution Modelling of Coronae and Winds in Solar-type Stars with Varying Rotation Rates. II. Stellar Winds*, under review.

In Preparation

Chen, Y.-H., et al., *First Global 3D MHD–Driven Modeling of Stellar Energetic Particle Events in Young Solar-type Stars*, in prep.

Conference Presentations

Aug 2026 COSPAR 2026, Florence, Italy.

Aug 2026	Magnetic Eclipse, Valencia, Spain.
Jun 2026	Cool Stars 23, Tokyo, Japan. *(Awarded Best Poster, granted oral presentation opportunity.)*
Mar 2026	89th DPG Annual Meeting & SMuK Spring Conference, Extraterrestrial Physics (EP) Division. FAU Erlangen-Nürnberg, Germany.
Aug 2025	16th Zhang Heng Symposium / Joint Solar–Stellar Physics Workshop (Centenary of Chinese Hand-Drawn Sunspot Records). Qingdao, China.
Jul 2025	IAU Symposium 400, "Solar and Stellar Multi-Scale Activity." EAFIT University, Medellín, Colombia.
May 2025	"Stellar X-Ray Flares in the Einstein Probe Era" Workshop (EP Science Center & NAOC Solar Magnetic Activity Group). Beijing, China.
Apr 2025	2nd International Conference of Deep Space Sciences (ICDSS 2025). Hefei, China. *(Best Presentation Award.)*
Jul 2024	45th COSPAR Scientific Assembly. BEXCO, Busan, Korea.
May 2024	IAU Symposium 388, "Solar and Stellar Coronal Mass Ejections." Jagiellonian University Observatory, Kraków, Poland.
2023	Annual Meeting of the Chinese Solar Physics Community. Chengdu, China.

Grants & Funding

Apr 2024 – May 2026	**Formation and Evolution of Stellar Coronal Mass Ejections**. *Principal Investigator*. Postgraduate Research & Practice Innovation Program of Jiangsu Province.
Sep 2024 – Aug 2027	**Lower-Atmospheric Signatures of Coronal Mass Ejection Initiation**. *Co-Investigator* (numerical simulation). Natural Science Foundation of Jiangsu Province (Youth Program). PI: Dr. Chen Xing.

Teaching

TA	*Introduction to Space Physics* (undergraduate course), Nanjing University.

Skills

Programming	Python, IDL, Fortran
Simulation codes	MPI-AMRVAC, SWMF
Methods	MHD simulation; solar and stellar data analysis

Languages

Mandarin Chinese	Native
English	Working proficiency

References

Ph.D. advisor	Prof. Xin Cheng, School of Astronomy and Space Science, Nanjing University.
Visit host	Dr. Julián D. Alvarado-Gómez, Leibniz Institute for Astrophysics Potsdam (AIP).
M.Sc. advisor	Assoc. Prof. Yu Dai, School of Astronomy and Space Science, Nanjing University.